

Call for Participation in the First Embodied Audition and Reasoning (EAR-1) Challenge

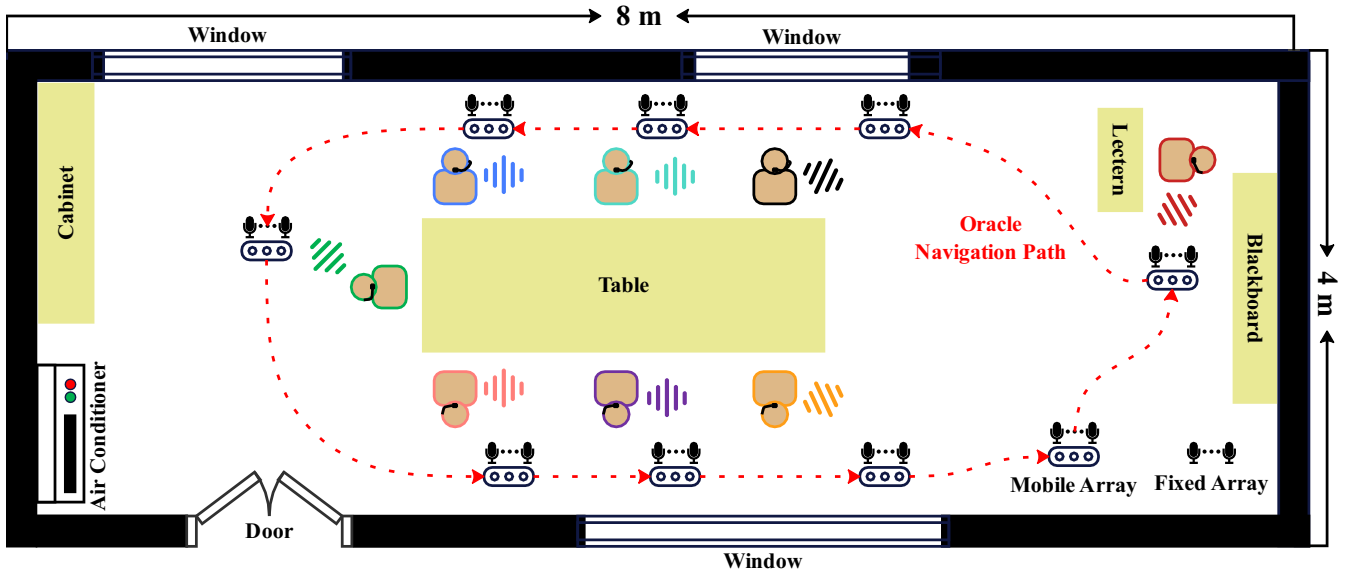


Fig. 1. Embodied speech perception under oracle navigation in Multi-Party Conversations. Paired fixed and mobile recordings support controlled analysis of the acoustic value of movement for speaker diarization and speech recognition.

Conventional multi-party speech systems observe conversations from a fixed far-field position, making them vulnerable to distant speech, background noise, reverberation, and speech overlap. Auxiliary cues, such as enrolled-speaker information and video, can improve robustness but require prior knowledge or additional sensors, making it difficult to isolate the value of changing the acoustic viewpoint.

Embodied devices, including robots, smart glasses, and wearable assistants, can actively change their acoustic perspective by rotating toward active speakers or approaching sound sources. This naturally leads to the idea of embodied speech perception, where the recording device moves to obtain more informative acoustic observations. Before considering how such movement should be planned, however, a fundamental question must first be answered: *if navigation is executed correctly, to what extent can movement itself improve speech perception performance in multi-party conversations?*

Accordingly, **the first Embodied Audition and Reasoning (EAR-1) Challenge**, themed **Embodied Speech Perception under Oracle Navigation in Multi-Party Conversations**, evaluates the impact of changing acoustic viewpoints under controlled oracle navigation. As illustrated in Fig. 1, EAR-1 provides 50+ hours of synchronized paired multi-channel recordings of the same multi-party meetings from fixed and moving microphone arrays. The moving array is carried by a human proxy following an oracle-guided path, while participants receive audio only without navigation trajectories, pose, video, or inertial information.

Specifically, EAR-1 comprises two tasks, each of which includes a *Fixed-Array Track* and a *Moving-Array Track*:

- **Embodied Speaker Diarization with Oracle Navi-**

gation (ESD-ON) requires systems to assign speaker labels to continuous multi-channel recordings with oracle-provided speech activity regions and overlapping speech. Performance is evaluated using diarization error rate (DER).

- **Embodied Speech Recognition with Oracle Navigation (ESR-ON)** requires systems to generate speaker-attributed transcriptions from continuous, unsegmented multi-channel recordings without provided speech boundaries. Performance is evaluated using concatenated minimum-permutation word error rate (cpWER).

Links:

- **Official website:** <https://coalboss.github.io/ear1>
- **Register form:** <https://forms.gle/oBDTLZCdumGLZpCA>
- **Contact email:** ear.challenge.office@gmail.com

Key Dates:

- **Sep. 15, 2026:** Training/dev data and baselines released
- **Nov. 16, 2026:** Evaluation data released
- **Dec. 18, 2026:** Final rankings and top-team invitations

Organizers:

- **Hang Chen**, USTC, China
- **Jun Du**, USTC, China
- **Mengyao Zhu**, Soochow University, China
- **Hui Bu**, AISHELL, China
- **Xin Xu**, AISHELL, China
- **Sabato Marco Siniscalchi**, University of Palermo, Italy
- **Kai Chen**, DeepCybo, China
- **Jia Pan**, iFLYTEK, China
- **Yanhui Tu**, Anhui University, China
- **Qiang Zhou**, AISPEECH, China

I. CHALLENGE DESCRIPTION AND SCIENTIFIC MOTIVATION

Conventional multi-party speech systems typically observe conversations from a fixed far-field position, making them vulnerable to distant speech, background noise, reverberation, and severe speech overlap. These challenges are common in real-world applications such as smart homes, meeting assistants, and human-robot interaction, where multiple speakers interact naturally without strict turn-taking. Although significant progress has been achieved in speech enhancement [1], [2], [3], speaker diarization [4], [5], and robust automatic speech recognition [6] through large-scale dataset [7], [8], [9] and advanced neural architectures [10], performance remains limited in highly dynamic conversational environments. For example, existing benchmarks such as CHiME-6 [11] still exhibit substantial recognition errors in realistic dinner-party scenarios, indicating that improving computational models alone may not fully address the limitations imposed by the acoustic observation itself.

Auxiliary modalities, including enrolled-speaker information [12], visual signals [13], and spatial cues [14], can further improve speech perception. However, these approaches typically require prior knowledge of target speakers or additional sensing modalities, making it difficult to isolate the fundamental value of obtaining a more informative acoustic viewpoint. In contrast, embodied devices [15], including robots, smart glasses, and wearable assistants, provide a new possibility: actively changing the position and orientation of the listening device to acquire more favorable acoustic observations. Changing the acoustic viewpoint alters source-to-array distance, direct-to-reverberant ratio, relative directions of arrival, and spatial separability of concurrent speakers, potentially providing information unavailable to stationary systems.

However, movement itself also introduces new challenges, including self-noise and rapidly varying acoustic transfer functions, which may affect spatial filtering, speaker tracking, and speech recognition. It therefore remains unclear whether, and under what conditions, the benefits of an adaptive acoustic viewpoint outweigh the additional difficulties introduced by movement. Before exploring more complex scenarios involving jointly learned perception and navigation, a fundamental question must first be answered: *when navigation is executed correctly, what is the intrinsic value of an embodied acoustic viewpoint for multi-party speech perception?*

Answering this question requires separating the acoustic effects of movement from errors in sound localization, path planning, and motion execution. Existing evaluations of embodied audition often couple these components within a single system. Consequently, a perception failure may arise from inaccurate localization, ineffective navigation, imperfect motion control, or limitations of the underlying speech model. Such coupled evaluation makes it difficult to determine whether changing the acoustic viewpoint itself benefits multi-party speech perception.

To isolate this factor, **the first Embodied Audition and Reasoning (EAR-1) Challenge**, themed **Embodied Speech**

TABLE I
PRELIMINARY STATISTICS OF THE EAR-1 CORPUS.

Dataset	Train	Dev	Eval	Total
Duration (h)	46	2	2	50
Sessions	33	2	3	38
Rooms	12	2	2	16
Participants	95	8	12	115
Male	48	4	6	58
Female	47	4	6	57

Perception under Oracle Navigation in Multi-Party Conversations, adopts an *oracle-navigation* setting. During data collection, a human proxy carries the recording device and correctly executes a predefined navigation policy without participating in the conversation. The term “oracle” refers only to the physical execution of navigation and does not provide participating systems with privileged acoustic labels or navigation metadata. This controlled setting removes navigation errors from the evaluation and allows the challenge to focus specifically on speech perception under correctly executed movement.

EAR-1 investigates this scientific question through synchronized, paired observations from fixed and moving microphone arrays, enabling controlled within-conversation comparisons between conventional fixed far-field perception and embodied speech perception. In doing so, EAR-1 aims to advance robust speech processing under dynamic acoustic conditions and establish a controlled benchmark for quantifying the value of an embodied acoustic viewpoint. It also provides a foundation for future challenge editions in which systems may jointly perceive conversational environments, reason about informative acoustic viewpoints, and learn navigation policies for improving speech perception.

II. DATASET AND RECORDING PARADIGM

As shown in Table I, the EAR-1 corpus is expected to contain approximately 50 hours of real multi-party conversations recorded in 38 sessions across 16 rooms. Each session provides synchronized, paired multi-channel audio recordings from a fixed microphone array and a moving microphone array. The reported duration refers to the underlying conversation time rather than the sum of the two recording conditions. The corpus will be divided into training, development, and evaluation sets, with speakers and rooms kept disjoint across the three subsets to support evaluation under unseen speakers and acoustic environments.

During recording, the fixed microphone array remains at a reference position, while a human proxy carries the moving microphone array and follows a predefined oracle-navigation path. The oracle-navigation policy is designed to provide controlled and reproducible changes in acoustic viewpoints rather than optimize speech perception through an adaptive model. Specifically, the navigation path follows three principles:

- 1) **Predefined trajectory:** The overall movement pattern is specified before recording, with deterministic paths such as clockwise or counter-clockwise exploration around the conversation area. This ensures consistent spatial

coverage and avoids uncontrolled motion behaviors across sessions.

- 2) **Active-speaker-oriented movement:** The human proxy follows an intuitive acoustic interaction principle by moving toward the direction with stronger perceived speech energy. This approximates how an embodied listener naturally approaches a more informative acoustic viewpoint while maintaining a predefined navigation constraint.
- 3) **Controlled random exploration:** Randomized movements are introduced within the predefined trajectory to increase diversity in acoustic viewpoints and prevent the benchmark from being limited to a single deterministic movement pattern.

The resulting trajectories provide physically executed viewpoint changes while avoiding the confounding effects of imperfect navigation. Participants are not provided with the navigation policy, trajectories, or device poses, ensuring that systems must rely solely on the recorded acoustic observations. The training and development sets will include speaker activity annotations and speaker-attributed reference transcriptions for both recording conditions. For **Embodied Speaker Diarization with Oracle Navigation (ESD-ON)**, oracle voice activity detection (VAD) regions will be provided for all subsets, including evaluation. For **Embodied Speech Recognition with Oracle Navigation (ESR-ON)**, systems will directly process continuous recordings without provided speech boundaries. Evaluation references, including speaker labels and transcriptions, will remain private and will be accessed only through the official evaluation server.

Close-talking microphones and auxiliary sensors, including video and inertial measurements, may be used internally for synchronization, annotation, and quality control, but will not be released as challenge inputs. Participants will receive only the multi-channel audio recordings from the fixed and moving microphone arrays. The two recording conditions correspond to the *Fixed-Array Track* and *Moving-Array Track*, respectively, and will be evaluated separately using the same task protocols.

III. TASKS AND EVALUATION

EAR-1 evaluates two fundamental capabilities for embodied speech perception in multi-party conversations: speaker diarization and speaker-attributed speech recognition. Each task contains two recording-condition tracks: a *Fixed-Array Track*, using audio captured by the stationary microphone array, and a *Moving-Array Track*, using audio captured by the moving microphone array following the oracle-navigation policy. The two tracks share the same conversations and evaluation protocols, enabling controlled comparison between conventional fixed far-field perception and embodied acoustic viewpoints.

A. Embodied Speaker Diarization with Oracle Navigation

ESD-ON addresses the “who spoke when” problem in continuous multi-channel recordings. For this task, oracle

voice activity detection (VAD) regions are provided, indicating speech-active intervals without revealing the number or identities of active speakers. Systems are required to estimate speaker activities, maintain consistent speaker identities throughout each session, and assign one or more speaker labels to regions containing overlapping speech.

Diarization error rate (DER) is adopted as the evaluation criterion:

$$\text{DER} = \frac{T_{\text{SC}} + T_{\text{FA}} + T_{\text{MD}}}{T_{\text{total}}}, \quad (1)$$

where T_{SC} , T_{FA} , and T_{MD} denote the durations of speaker confusion, false alarm, and missed detection, respectively, and T_{total} is the total reference speaker time. Since oracle VAD is provided, DER mainly reflects speaker confusion and overlap attribution rather than speech activity detection errors. No forgiveness collar will be applied, and overlapping speech will be included in evaluation. Lower DER indicates better performance.

B. Embodied Speech Recognition with Oracle Navigation

ESR-ON evaluates speaker-attributed speech recognition from continuous, unsegmented multi-channel recordings. Unlike ESD-ON, no speech boundaries or activity annotations are provided. Systems must jointly handle speech activity detection, speaker attribution, and speech recognition, and generate speaker-indexed transcripts with temporal information.

For each session, recognized words assigned to each speaker are concatenated in temporal order. Concatenated minimum-permutation word error rate (cpWER) is then computed over all possible mappings between reference and hypothesized speakers:

$$\text{cpWER} = \min_{\pi \in \Pi} \frac{N_{\text{Sub}}^{(\pi)} + N_{\text{Del}}^{(\pi)} + N_{\text{Ins}}^{(\pi)}}{N_{\text{ref}}} \times 100\%, \quad (2)$$

where Π denotes the set of possible mappings between reference and hypothesized speakers, and $N_{\text{Sub}}^{(\pi)}$, $N_{\text{Del}}^{(\pi)}$, and $N_{\text{Ins}}^{(\pi)}$ represent the numbers of substitutions, deletions, and insertions under the corresponding speaker assignment, lower cpWER indicates better performance.

C. Ranking and Evaluation Protocol

Separate leaderboards will be maintained for ESD-ON and ESR-ON under both fixed and moving recording conditions. The official ranking metrics are DER for ESD-ON and cpWER for ESR-ON. All metrics are computed independently for the *Fixed-Array Track* and *Moving-Array Track*, allowing participants to quantify the impact of changing acoustic viewpoints.

Reproducible baseline systems will be released for both tasks, together with official output formats, scoring scripts, and evaluation configurations. During the challenge period, organizers will provide technical documentation, an FAQ, and communication channels to support participants. Final submissions will be evaluated against hidden references using the official scoring scripts and fixed configurations. Participants will submit a system-description report documenting model architectures, training resources, external data usage, and decoding strategies. Final rankings will be announced after submission validation and compliance checks.

IV. GUIDELINES FOR PARTICIPANTS

Q1 Can external data or pretrained models be used?

R1 Publicly available external datasets and pretrained models are allowed. All external resources must be cited, accessible to other participants, and reported in the system-description report. No additional resource may be introduced after the release of evaluation data.

Q2 What information will be provided?

R2 Participants will receive officially released audio recordings and metadata specified by each task. Speaker labels and reference transcriptions will be available for training and development sets, while evaluation references will remain hidden. For ESD-ON, oracle VAD information will be provided as defined in the task protocol.

Q3 What are the submission requirements?

R3 Teams may participate in one or more leaderboards. Each submission must contain system outputs in the official format and a system-description report covering model architecture, training procedure, external resources, and decoding configurations. Unreleased information, including navigation trajectories, device poses, auxiliary sensors, and organizer-only annotations, cannot be used as system inputs. Final rankings are subject to organizer compliance checks.

V. FULL CHALLENGE SCHEDULE

- **Aug. 14, 2026:** Challenge website, rules, and registration open
- **Sep. 15, 2026:** Training/development data and baseline systems released
- **Oct. 14, 2026:** Development leaderboard opens
- **Nov. 16, 2026:** Evaluation data released
- **Dec. 7, 2026:** Final system outputs due
- **Dec. 14, 2026:** System-description reports due
- **Dec. 18, 2026:** Final rankings and top-team invitations announced
- **Jan. 7, 2027:** Invited 2-page papers due
- **Jan. 21, 2027:** Paper acceptance notification
- **Jan. 28, 2027:** Camera-ready 2-page papers due
- **May 16–21, 2027:** EAR-1 session at ICASSP 2027, Toronto, Canada

The public leaderboard phase runs for approximately two months from its opening to the announcement of the final rankings. Up to five top-ranked teams will be invited to submit 2-page papers and present their systems in person at ICASSP 2027. The EAR-1 session will include an organizer overview, the winner announcement, presentations from the invited teams, and an open discussion on future directions in embodied audition and reasoning.

VI. LIST OF POTENTIAL PARTICIPANTS

The following researchers have been identified as potential participants:

- Duke Kunshan University (China), Ming Li (ming.li369@duke.edu)
- Institute of Acoustics, CAS (China), Qiang Fu (fuqiang@mail.ioa.ac.cn)

- Nanjing University (China), Jing Liu (lujing@nju.edu.cn)
- Kyoto University (Japan), Tatsuya Kawahara (kawahara@i.kyoto-u.ac.jp)
- Northwestern Polytechnical University (China), Lei Xie (lxie@nwpu.edu.cn)
- Nanyang Technological University (Singapore), Eng-Siong Chng (aseschng@ntu.edu.sg)
- Shanghai Jiao Tong University (China), Yanmin Qian (qian-ym@cs.sjtu.edu.cn)
- Tsinghua University (China), Wei-Qiang Zhang (wqzhang@tsinghua.edu.cn)
- The Chinese University of Hong Kong, Shenzhen (China), Haizhou Li (haizhouli@cuhk.edu.cn)
- University of Rochester (USA), Zhiyao Duan (zhiyao.duan@rochester.edu)
- Xiamen University (China), Qingyang Hong (qyhong@xmu.edu.cn)
- Sogang University (Korea), YeonJin Kim (yjkim@sogang.ac.kr)
- Zhejiang Lab (China), Yongxiong Xiao (xiaoyx@zhejianglab.com)

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